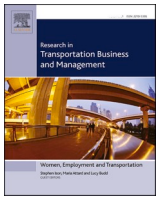
Research in Transportation Business & Management 48 (2023) 100909



Contents lists available at ScienceDirect

Research in Transportation Business & Management

journal homepage: www.elsevier.com/locate/rtbm





Boosting intermodal rail for decarbonizing freight transport on Java, Indonesia: A model-based policy impact assessment

Ronald A. Halim *

Equitable Maritime Consulting, Jan van Riebeekstraat 68, 2595, WS, the Hague, the Netherlands

ARTICLE INFO

Keywords:
Modal shift policy
Intermodal transport
Decarbonization
Transport model
GHG emission reduction
Indonesia
Rail freight

ABSTRACT

This paper assesses the impact of a modal shift action program aimed at reducing CO_2 emissions from surface freight transport on Java - Indonesia's center of economy and population. The action program combines fiscal measures and infrastructure development strategies which are designed through consultation with policy makers and relevant stakeholders. I test three ambition levels of the action program -basic, moderate, strong on CO_2 emissions, using a 4-step freight transport demand model and a fleet emission model. For each of the ambition levels, I estimate the shift from road to rail freight and savings in CO_2 emissions attained. I found that a significant transport demand could be shifted from road to rail where rail freight share could reach up to 20% of the total cargo transported by 2030. This shift to rail will also lead to a considerable reduction in the total annual CO_2 emissions, up to 21.6% under the strong ambition scenario. In terms of economic benefits, the shift will also reduce total transport costs up to 7.6% under the most ambitious scenario.The high effectiveness of the measures to abate emissions substantiates the need for a comprehensive cost-benefit analysis to further assess the cost-effectiveness of the action program. Based on the modelling, several following key policy measures are identified to improve the share of rail freight: (1) improving service quality of rail freight, (2) building sufficient rail infrastructure capacity, and (3) setting up dedicated funding program for intermodal rail infrastructure.

1. Introduction

Indonesia, the fourth most populous country in the world with a population of over 257 million, is the largest member state of the Association of Southeast Asian Nations (ASEAN) and among the countries with the largest economies and fastest economic growth in the world. Java island is Indonesia's center of population with more than150 million inhabitants (BPS, 2018b) and the backbone of Indonesia's economy, with a 60% share of Indonesia's national GDP (BPS, 2018a). It is also the island with the highest GHG emissions in Indonesia. In 2015, the CO_2 emission from domestic freight transport on Java −11.53 MTonne is estimated to compose around 8.12% of the total CO_2 from Indonesia's transport sector.[1] This can be attributed to the use of road freight as the dominant transport mode on Java and the rest of Indonesia with a mode share of >90% (Mulyono, 2019).

Based on the upward trend of the GDRP of Java's biggest provinces in the past decades (Fig. 1), a high future growth of freight transport activity is likely. Without mitigation action, this trend will significantly increase transport emissions. By 2030, carbon emissions of domestic freight transport on Java might reach 17.36 MTonne (GIZ, 2021) if no measures are taken.

The combination of road and rail transport ('intermodal') is a transport alternative that has a high potential to help reduce logistics costs and GHG emissions simultaneously. Specifically, rail transport has lower cost and fuel intensity compared to road transport. Java island with legacy railway infrastructure which stretches from West to East Java with around 3783 km, has a potential to attain modal shift from road to rail transport. The rail networks consist of two main lines namely the North Lines and South Lines which capacity is currently being expanded into double-tracked rail lines. The lines could be extended to connect the big ports in the west of Java (Tanjung Priok Port), Tanjung Emas Port in Central Java, and Tanjung Perak Port in east of Java. Fig. 2 shows the railway lines (single and double track) and the main road on the island of Java. These railway lines have the characteristics where they can support long-haul distance such as those between Jakarta and Surabaya which are in the range of 720 km. They therefore meet

* Corresponding author.
E-mail address: ronald@equitablemaritimeconsulting.nl.

[1] Based on the output of modelling exercise in this study and the total emission from transport sector reported in the Indonesia Biennial Update Report to UNFCCC 2018.

https://doi.org/10.1016/j.rtbm.2022.100909
Received 31 January 2022; Received in revised form 31 October 2022; Accepted 1 November 2022
Available online 18 November 2022
2210-5395/© 2022 Elsevier Ltd. All rights reserved.

framework conditions that could promote modal shift when there is sufficient rail freight capacity and commitment from the government.

However, most of the companies who ship their goods across Java ('shippers') currently do not consider rail freight as an attractive mode of transport compared to trucking. This is reflected by the very low modal share of rail freight on Java (in tonne-kilometers) which has been estimated to be around 1.74% in 2020 (GIZ, 2021). A recent study conducted by conducted by the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ, 2021) has identified the causes of low rail modal share. They are (1) uncompetitive costs and performance compared to road transport, namely higher costs, and longer transit time of intermodal rail freight; (2) poor service quality which affects the risk and safety of goods delivery. The root causes of those three main factors are the lack of adequate infrastructure for intermodal services, the lack of market competition on intermodal rail sector, and the lack of prioritization for rail freight services compared to passenger transport. As a result of these problems, freight transport on Java is dominated by road transport and rail infrastructure is mainly used to serve passenger transport rather than freight transport.

The low rail modal share on Java also indicates a prominent gap between the current situation and the goal set in the National Railway Masterplan (RIPNAS) 2018. In this masterplan, rail modal share is targeted to reach 11–13% by 2030. If no long-term measures are taken to help shift the demand for freight transport to rail and to maritime freight, Indonesia will face a huge challenge to decarbonize its freight transport system in the future. Reliance on road transport will be very difficult to break and the costs of building rail infrastructure needed to increase the competitiveness of rail transport will be substantially higher.

To respond to the transport decarbonization challenge in Indonesia, I assess a set of measures that are aimed to shift the demand from road to rail transport. This set of measures has been elaborated by a technical design study conducted by the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ, 2021)in cooperation with the Ministry of Transportation (MoT's DG Land Transportation, DG Railway) under the TRANSfer III project with a consultation process involving multiple stakeholders from government agencies and from the freight transport sector, and with funding from the International Climate Initiative of Germany's Federal Ministry for the Environment, Nature Conservation and Nuclear Safety. This paper presents the underlying modelling and scenario development work which has been used to assess the effectiveness of these decarbonization measures as reported by Halim, Breemersch, Nahry, and Kusuma (2021). The summary of results of the technical design study are formulated as an Intermodal Action Programme as depicted in Fig. 3. The measures designed in the action programme are comprised of technical, institutional and regulatory measures which share common characteristics and directions with a list of policy instruments discussed in the World Bank report by Aritua (2019).

While there are many potential mitigation measures, the impact of these measures and their combination on the reduction of GHG emissions is unknown. Consequently, it is also difficult to understand the levels of carbon emission reduction attainable and what ambition levels, including its investment costs are needed to achieve desired CO_2 abatement. Therefore, an analytical approach is needed to assess the impact of these measures on GHG emission reduction. This is important to help guide and prioritize the adoption of specific action programme and to set a realistic mitigation target for freight transport sector, especially in developing countries which share similar characteristics and challenges as Indonesia.

Unfortunately, there have not been any in-depth quantitative studies that assess the required policy measures to support the realization of the decarbonization target for Indonesia. An ambitious target will generally require massive and rapid changes, often involving capital investments that might not have clear prospects for profitability. This could bring considerable economical disruptions (i.e., loss of profit) to industry stakeholders, notably shippers, logistics service providers, and national governments. Without appropriate national policies that can provide strong incentives and mechanisms that favor the shift to rail, ambitious targets and strict regulations might face strong resistance by relevant industry stakeholders.

This paper presents a systematic assessment of measures designed to shift freight transport demand from road to rail and its implications for the required supporting policy measures until 2030. It does not look into actions to improve the energy consumption of rail itself (e.g. via electrification) or to make freight trips shorter or unnecessary ('avoid'). I establish a modelling framework that consists of java freight transport and emission models to study the impact of mitigation measures on CO_2 emissions. I project the transport demand for Java until 2030 and estimate the modal share of road and rail. Next, I test the impact of intervention measures which cover technical, financial, and institutional policies on modal shift and GHG emissions of freight transport on Java.

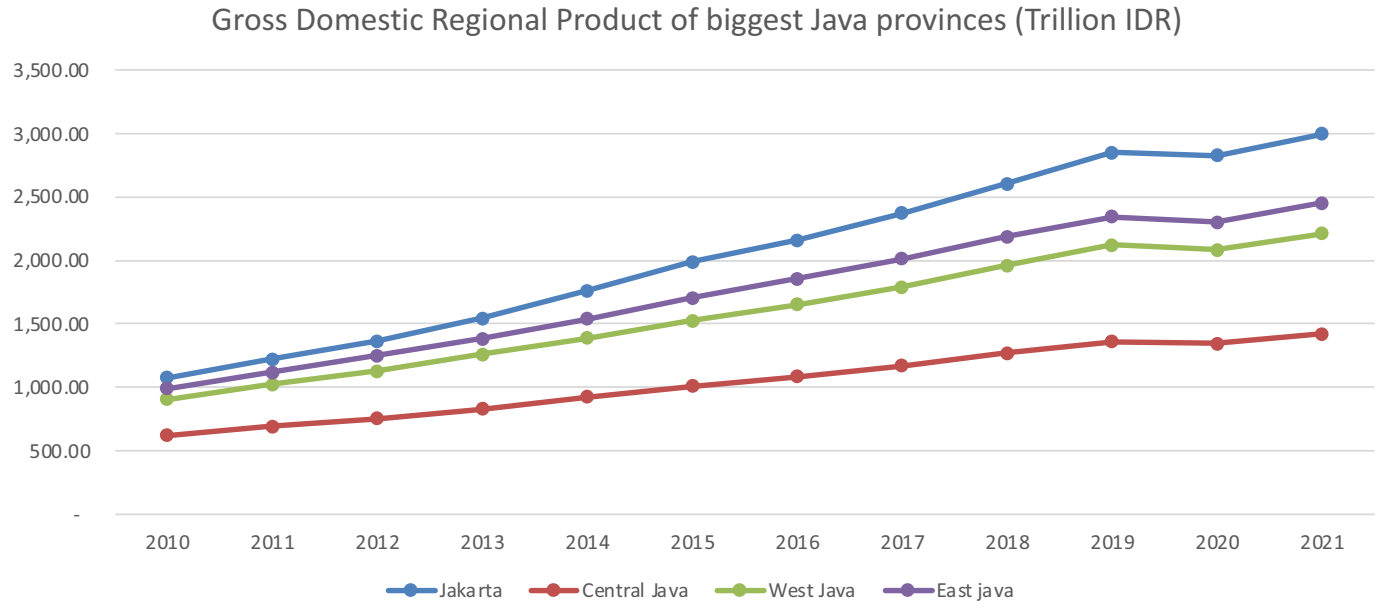


Fig. 1. Trend of Java's GDP, Source: (BPS, 2018a).



Fig. 2. Railways and Main Road on Java Island.
Esri, HERE, Germin, FAO, NOAA, USGG, Open StreetMap contributors, and the GIS User Community, 2020.

The contribution of this study is two folds. First, it assesses the impact of policy measures to shift freight from road to rail on the abatement of GHG emission from freight transport in a developing country such as Indonesia.[2] Second, based on the quantitative impact assessment, it provides concrete policy recommendations that can be implemented by relevant government agencies. The remainder of this paper is structured as follows. In Section 2, I review the available literature focusing on modal shift strategy to reduce GHG emissions. Section 3 presents the methodology used to assess GHG reduction potential of different measures. In Section 4, I describe GHG reduction scenarios that are simulated using the modelling framework. Section 5 presents the results of these scenarios. Section 6 provides discussions on the policy implications derived from the results and Section 7 summarizes the scholarly contribution of this study.

2. Literature review

Given the global pattern of the reliance of freight transport on road mode, shift to intermodal rail from road freight is considered to be an effective strategy to curb GHG emissions. This is primarily because rail transport is generally less carbon intensive compared to road freight (IEA, 2019; Kaack, Vaishnav, Morgan, Azevedo, & Rai, 2018; Pinto, Mistage, Bilotta, & Helmers, 2018). Another contributing factor is also because a large fraction of rail system in the world is powered by electricity (Kaack et al., 2018) which makes the potential for deep decarbonization to be more attainable and dependent on the use of green electricity. While the majority of Indonesia's rail network is still powered by fossil fuel, the Indonesian government has planned to expand the electrification of the existing national rail network while aiming to build electrified rail network for new rail connections such as high-speed rail connection from Jakarta to Bandung.

A report from IFEU (Institut fur Energie- und Umweltforschung Heidelberg) shows that intermodal transport in Europe on average consumes 16% less energy than road transport (IFEU & Sgkv, 2009). Furthermore, Craig, Blanco, and Sheffi (2013) found carbon intensity from intermodal transport is on average 46% lower than road transport. Although the research also notes that there is a large variance in the distribution of carbon intensities with several cases where intermodal rail transport is more carbon intensive than truck transport due to high GHG emissions from terminal operations.

Recently studies that focus on the impact of modal shift as a strategy to reduce emissions in developing countries have started to emerge. Among others, they include studies conducted for China, India, and South Africa. Chen, Wu, and Zong (2020) analyses the impact of modal shift policy on China's carbon emission from freight transport and the country's economy. The author found that modal shift policy to rail has the potential of reducing up to 17% of carbon emissions compared to do nothing scenario. This positive environmental benefit is however not always accompanied by economic benefit in cases where uniform policy is applied without paying attention to specific characteristics and strengths unique to regions. Furthermore, for India, CAT (2020) showed the massive potential of CO_2 emission reduction – (i.e. 42% reduction in 2050) that can be achieved with modal shift to rail both from passenger and freight. However, it is not clear how much CO_2 reduction can be attained from shifting road freight to intermodal rail transport. Another study for India also confirmed this potential for decarbonizing freight through modal shift policy (Grütter, 2015). A study conducted by Department of Environmental Affair of South Africa indicated that freight shift from road to rail could reduce 3 $MtCO_2eq$ emissions from freight transport and contributes to 0.66% of the total mitigation potential in South Africa. However, this policy may also cause negative short-term impacts on labor employment and economies in the country.

Unlike other developing countries, studies about the impact of modal shift on CO_2 emissions in Indonesia are limited and not focused on decarbonization strategy. Most of the studies on modal shift are aimed to assess the efficacy of a policy measure to shift road freight to rail freight for improving Indonesia's logistics performances and economies such as exemplified in (de Baat, den Hertog, de Jong, de Regt, & Wijgergangs, 2015; Hartono, Mochtar, & Widyastuti, 2017; Norojono & Young, 2003). A notable exception is the study of Nugroho, Whiteing, and de Jong (2015) who estimated carbon emission reduction that can be attained through modal shift on Java. However, the focus of the study is to test 5 policy measures to shift container exports from road to rail from 16 regions on Java, through 4 main ports, using 2 alternative modes (road and rail). These policy measures represent reduction in rail travel time, reduction in rail tariff, as well as increase in time and costs of road mode. In the previous work of Nugroho et al. (2015) and Nugroho, Whiteing, and de Jong (2016) the policy measures are tested independently as individual policy scenarios. These studies, therefore, have not assessed the potential for modal shift for freight transport from and to all regions on Java, and the joint impact of all the measures on modal shift to rail transport.

In this study, I aim to contribute to the policy questions regarding the effectiveness of decarbonization measures through modal shift to rail in a developing country such as Indonesia. Within this context, the study adds to the literature of freight transport decarbonization in developing

[2] Technical Design Study of Action Programme on Intermodal Freight Transport on Java, Indonesia.

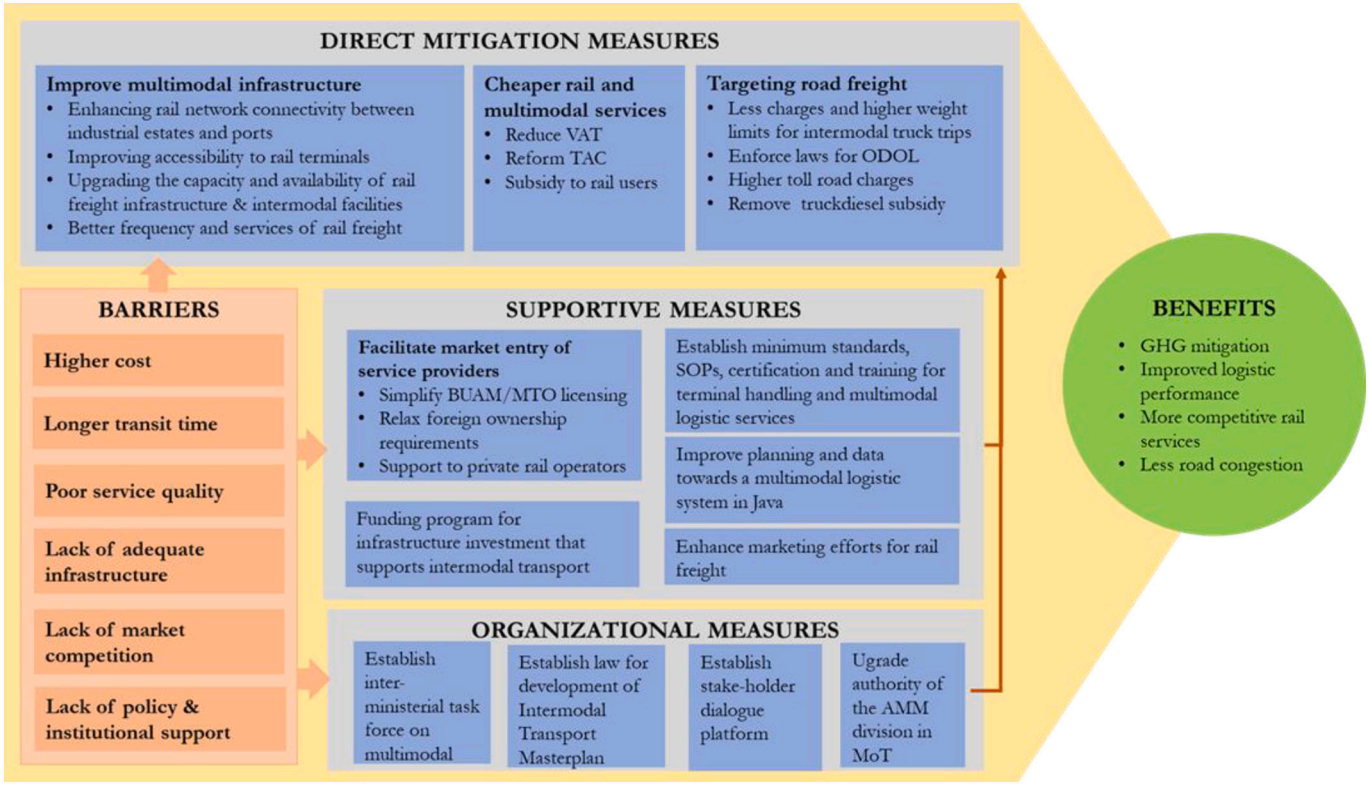


Fig. 3. Overview of proposed Intermodal Action Programme studied in the technical design study (GIZ, 2021).

countries in the following manner. The study firstly, assesses the joint impact of an action program which combines extensive policy measures aimed to promote modal shift to intermodal rail transport. These measures are results of a technical design study that was carried out through a consultation with policy makers and relevant stakeholders in Indonesia, namely Indonesia's Ministry of Transport and Association of Indonesian Freight Forwarder (GIZ, 2021).

Secondly, the model-based assessment covers broader geographical scope than what has been analyzed in the existing literature. Specifically, I model transport demand between all 118 districts on Java for all commodities which include both container and non-containerized cargoes. The model takes into account infrastructure limitations, routing behavior of the shippers, and allows the testing of broad policy measures that are not only limited to modal shift strategy but also currently implemented measures such as alternative fuel measures using biofuel, and minimization of over dimensioned and overloaded trucks.

Thirdly, the study provides specific insights on policy implications which can serve as a further guidance for Indonesia's government in developing a national transport decarbonization strategy through modal shift to rail. Although, the study is specific to Indonesia, I hope that the insights attained from this study could contribute as evidence and guidance in the adoption of modal shift policy in developing countries.

3. Modelling framework to assess the impact of GHG reduction measures

Estimation of the future CO_2 emission of freight transport on Java is performed using a freight transport demand model specifically built for Java island –the Java Freight Model (JFM) and the "ASIF" (Activity, Structure, Intensity, Emission Factor) method (Kooshian et al., 2017; Schipper, Marie-Lilliu, & Gorham, 1999). The ASIF model is a bottom up GHG emission model which enables the conversion of transport work in (tonne-kilometers) into GHG emissions. The model is based upon the following structure:

- Activity describes indicators for transport activity of freight or passenger
- Structure describes indicators for transport activity by modes, specified in Vehicle Kilo meter Travelled (VKT)
- Intensity describes variables that measure fuel or energy use by modes
- Emission Factor describes variables which explain the fuel mix and emission per unit of fuel by modes

The JFM is designed to project Java's freight transport activities (to project the A and S in the "ASIF") for all major transport modes (i.e. road, rail, sea) and routes while taking into account the decisions of different stakeholders (government, shippers, transport service providers) in a given transport network. The development of this model was motivated by the need to test the impact of detailed transport and economic policy measures and project the market share of freight transport modes with high degree of accuracy. Based on the projected activity (transport demand) and structure (modal share), the total fuel consumption for each mode, and vehicle type is computed using fuel intensity (I) data obtained from ICCT and the average load factor data for each vehicle type obtained from the Motor Vehicle Weighing Implementation Unit (UPPKB). Next, the total carbon emission is estimated based on the multiplication of each fuel type with emission factor data (EF). Data for emission factors is obtained from PPTMBG-Lemigas study (Novie, 2017).

The JFM model is built on the four-steps freight transportation modelling approach which typically includes freight trip generation, trip distribution, modal split, and traffic assignment (Ortúzar & Willumsen, 2011). It takes the trade flow from the 2011 national OD survey data as an input to estimate freight transport demand (in tonne-kilometer) until 2030 by modes and routes. The model has a spatial resolution defined at regional level (kabupaten level) which covers 118 origin and destination on Java. Fig. 4 provides the overview of the modelling framework. The modelling framework consists of the following components: (1) demand

projection model which includes estimation for trip generation and distribution, (2) mode choice model, (3) route choice model (4) fuel consumption model and (5) emission estimation model.

3.1. Gravity model to model distribution of transport demand

The projection for freight transport demand on Java is performed using a gravity model that takes into account relevant socio-economic variables such as (1) population of origin (2) population of destination (3) GDRP (Gross Domestic Regional Product) of origin (4) GDRP of destination (5) travel time between ODs. Moreover, based on the estimated relationship of the variables, I computed the freight volumes and compare it with the observed volume based on the 2011 OD data. The result of the comparison shows a good fit between the estimated and observed volume.

The model is specified at district (kabupaten) level which allows estimation of baseline and future freight flows between 118 origin and destination. There are five provinces considered in the model: DKI Jakarta (6 districts), Jawa Barat, (26 districts) Jawa Tengah (35 districts), DI Yogyakarta (5 districts), Jawa Timur (38 districts) and Banten (8 districts).

$$Ln(TD_{od}) = constant + GO.Ln\ GDRP_o + GD.Ln\ GDRP_d + PO.LnPop_o + LnPD.Pop_d + CoT.LnTT_{od} \quad (1)$$

where:

TDod = transport demand from origin o to destination d.
constant = average transport demand.
GO = coefficient of origin's GDRP.
GD = coefficient of destination's GDRP.
PO = coefficient of origin's population.
PD = coefficient of destination's population.
CoT = coefficient of travel time between OD.
GDRPo = Gross Domestic Regional Product of origin.

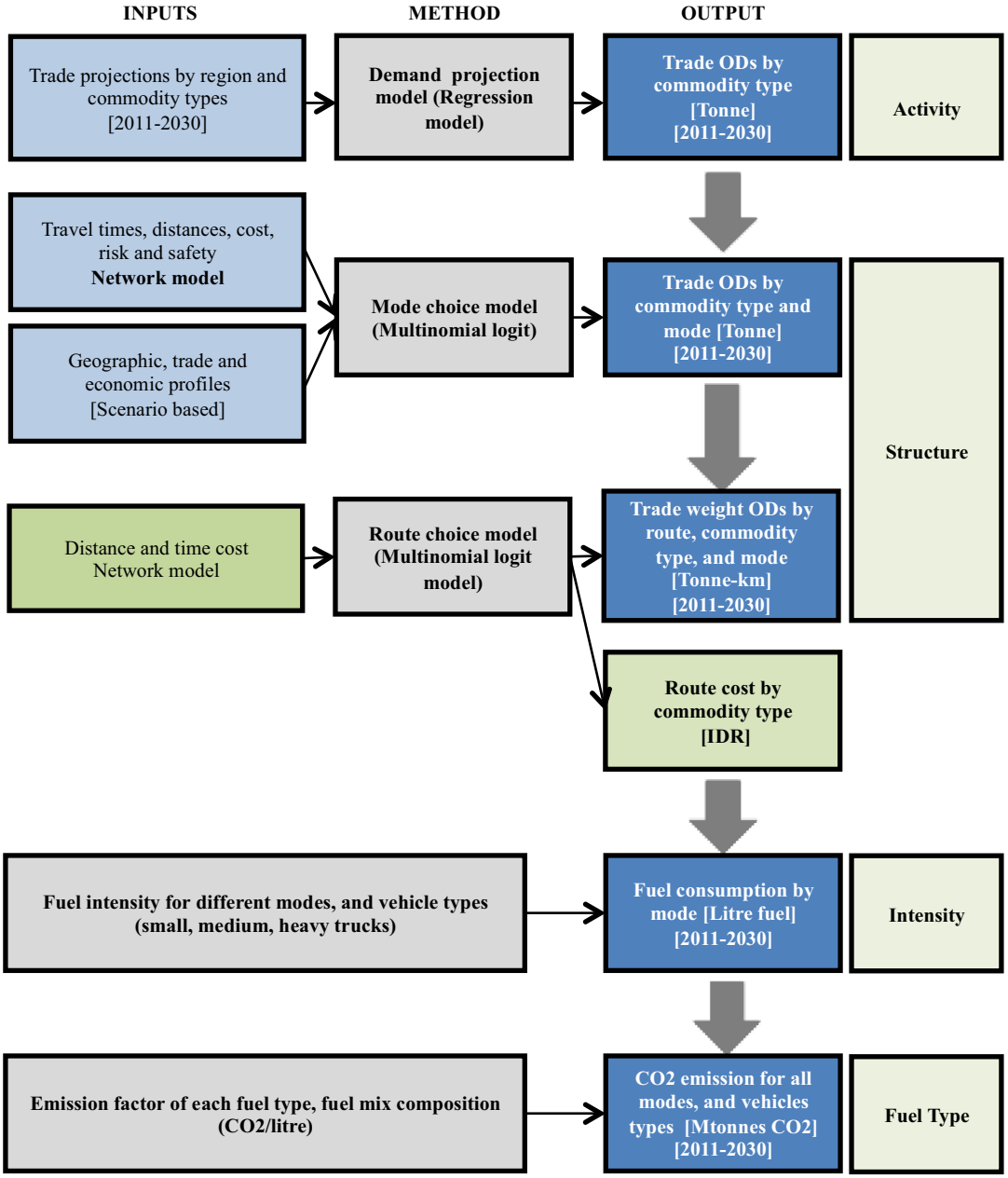


Fig. 4. Modelling framework for estimating impact of policies on freight transport demand and GHG emission on Java.

GDRPd = GDRP of destination.
Popo,d = Population of origin, destination.
TTod = Travel time between origin and destination.

To develop the transport demand scenario, I estimate the growth of GDRP and population. Furthermore, I estimate different growth patterns between both scenarios and estimate the trade volumes based on the coefficients of the model and the growth of all the variables. The detailed description and validation of the model can be found in the annex 1.

3.2. Transport cost and time schematization

Among different possible determinants, transport cost and time are typically the main determinants for modal choice by the shippers (Tavasszy & de Jong, 2014). In practice, there are a number of components that contribute to transport costs such as transloading costs, travel distance, travel time, and the value of time of the commodities. Hence to be able to reflect the impact of changes in one of these components on the overall transport cost from origin and destination, a detailed schematization and breakdown of transport costs component is necessary.

Transport costs for all origins and destinations were estimated by obtaining information about unit transport costs (IDR/t kilometer) from stakeholder interviews and further estimating these for all origins and destinations using transport network modelling. Transportation costs of a route is broken down into different components as shown in Fig. 5. In this way, the model could reflect scenarios where policy measures change specific components of transport costs such as transloading costs, or costs of shipment from and to rail terminals. As shown in Fig. 5, the cost model also takes into account the variation of costs associated with different alternative modes.

The formal definition of the cost model is delineated below:

$$C_r = \sum_{t \in r} A_t + \sum_{l \in r} c_l + \alpha \left(\sum_{t \in r} T_t + \sum_{l \in r} t_l \right) \tag{2}$$

where:
C_r = unit cost of route r from origin to destination (IDR/Tonne),
t = terminals used by the route,
l = links used by the route,
A_t = unit cost of transloading at terminal t (IDR/Tonne),
c_l = unit cost of transportation over link l (IDR/Tonne),
T_t = time spent during transhipment at terminal t (hour),
t_l = time spent during transportation over link l (hour),
α = value of transport time (IDR/h).

3.3. Mode choice model

In this study I developed an aggregate discrete choice model to analyze the impacts of policy measures on the preference of shippers regarding their choice of transport mode. The model's parameters are estimated based on an observation on shippers' preference with regard to the mode used. These observation data was obtained through a survey which records the volume of commodities transported between regions and the modes used to transport these commodities, also called the stated preference (SP) survey. The design of the survey is elaborated on the data section below. Using the survey data, a statistical estimation is performed to construct a mode-choice model that explains the preference of a shipper over available competing modes.

There are different types of discrete choice models that can be used to estimate modal share of transport modes. The most widely used models are logit and probit models (Ben-Akiva & Bierlaire, 1999). In this study, I invoked a Multinomial logit model for the purpose of estimating a mode choice model for freight transport on Java across 3 main modes: road, rail, and sea. This model is chosen because it is relatively simple and features a choice probability function that is easy to interpret. It is parsimonious yet able to capture choice behavior of shippers across multiple transport modes. Other possible model type such as mixed multinomial logit was not experimented due to limited time and budget. I experimented with several utility functions of the shippers, where attributes that are documented in the literature to be of major concerns for shippers are incorporated into model specification. Specifically, research carried out by Norojono and Young (2003) pointed out the relevance of transport costs, transport time, quality and flexibility for modal choice of freight transport on Java. Quality attribute can be expanded into more specific sub-attributes such as reliability, safety, truck condition, travel route, distance to rail, travel formation while flexibility includes the notion of frequency, responsiveness, and time departure. In order to retain the simplicity of the model, and the number of parameters that need to be estimated, the number of attributes incorporated in the model are limited to those which represent the most important factors that have been found in literature as studied by (Wanders, 2014). Furthermore, this limitation was also implemented to promote higher response rates among the respondents and to limit the complexity of the experimental design of the survey. The attributes incorporated in the final specification of the model are transport costs, travel time, on-time reliability, and risk and safety. The following model specifies the utility function of the shipper:

$$u_{odm} = asc_m + CF_{\mathrm{m}}.TC_{odm} + TF_m.TT_{odm} + RS.RnS_{od} \tag{3}$$

$$P_{m=} \frac{e^{u_{odm}}}{\sum_{m=1}^{M} e^{u_{odm}}} \tag{4}$$

where:
P_m = the choice probability of mode m,
$\mathrm{u_{od_m}}$ = the choice utility of mode m between origin o and destination d,
$\mathrm{asc_m}$ = alternative specific constant for mode m,
$\mathrm{CF_m}$ = transport cost coefficient for mode m.
$\mathrm{TC_{odm}}$ = transport cost for mode m between origin o and destination d,
$\mathrm{TF_m}$ = travel time coefficient for mode m.
$\mathrm{TT_{odm}}$ = travel time for mode m between origin o and destination d,
RS = risk and safety coefficient,
RnS_{od} = risk and safety variable between origin o and destination d,
RnS_{od} = [0, 5].

The estimation for the parameters of the multinomial logit model has been carried out using maximum likelihood estimation method which is provided by a software package Python Biogeme (Bierlaire, 2016). The estimation is done using SP data obtained from both a web-based survey and an in-person survey which was conducted in a multi-stakeholder workshop organized by GIZ on 12 February 2020 in Jakarta. The results of the estimation are presented in Table 1. The calibrated multinomial logit model has the adjusted likelihood value (Adjusted rho-square) of 0.524, reflecting satisfactory explanatory power of the model. All the mode attributes variables show the expected signs where transport costs, travel time have negative sign while risk and safety, and on-time reliability have positive signs. Most of the attributes are significant at 95% confidence level except for on time reliability which is found to be not significant. This result indicates that modal choice on Java is not sensitive to on time reliability of the freight shipments and shippers might have high tolerance on the punctuality of the cargo delivery services. This may be caused by the limited variation in on time reliability attribute presented in the survey which ranges from 70% to 90%. This implies that higher difference in on time reliability level is needed to increase the sensitivity of shippers towards this attribute.

When applying the model to predict rail volume using the estimated coefficient values, a discrepancy between the observed and estimated total rail volume is observed. Hence, after estimation based on the SP data, the mode choice model is further calibrated using observed rail freight volume in 2015 as the base year. This data is obtained from the

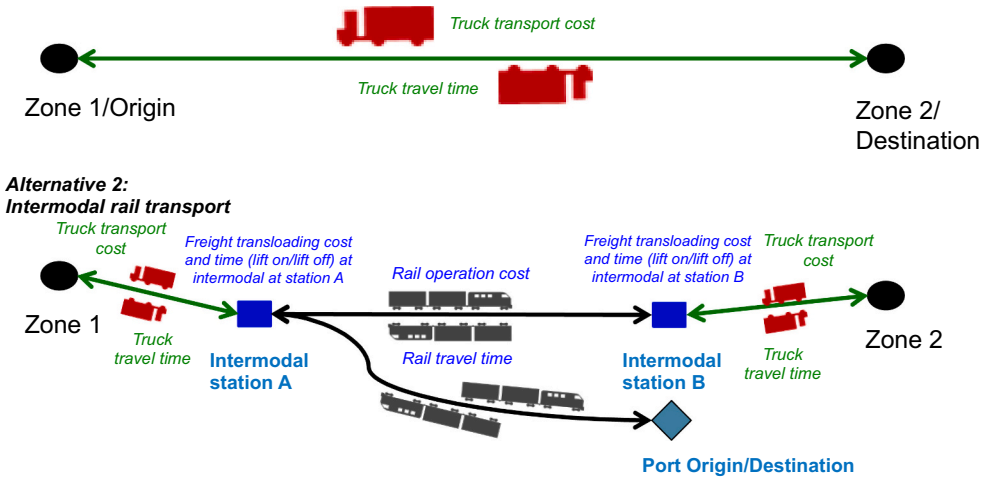


Fig. 5. Transport network schematization to estimate transport cost between origin and destination.

Table 1
Mode choice model estimation results.

Utility parameter	Coeff	Robust *t*-test	Robust *p* value
Alternative specific constant road	0	12.42	0.00
Alternative specific constant rail	−2.98 (−3.71)	−40.22	0.00
Alternative specific constant sea	1.36	−41.5	0.00
Transport cost road	−1.79	−1.96	0.00
Transport cost rail	−1.56	−2.31	0.00
Transport cost sea	−1.59	−1.44	0.00
Travel time road	−0.0374	−2.11	0.00
Travel time rail	−0.0825	−2.46	0.00
Travel time sea	−1.33	−1.52	0.00
Risk and safety	5.23E-01	−11.52	0.00
On time reliability *	2.39E-01	−0.21	0.12
Est. Parameters	10		
Adjusted rho-squared	0.524		
No. of observations	180		

* The variable on time reliability is not significant.

Directorate of Railway Traffic and Transport Data, Directorate General of Railway. In this process the coefficient of rail's alternative specific constants (ASC) was adjusted iteratively until the difference between the observed and predicted volume and tonne-kilometers of rail freight in 2015 is minimized. The adjusted rail ASC has the value of −3.75 which indicates that rail mode is the least attractive mode of transport due to the performance of in terms of transport costs, travel time, and risk and safety. The trend in the growth of rail freight volume has also been cross validated with the available statistics. Table 2 provides a comparison between modeled and reported freight volume and tonne-kilometers. While the modeled rail freight volume is similar to the reported statistics, there is a considerable discrepancy between modeled and reported rail tonne-kilometers. Upon an investigation on the possible sources of the errors, it becomes evident that this discrepancy is to a large extent, caused by differences between OD patterns estimated using eq. 1 and the OD data referred by the Indonesian ministry of transport such as what is presented in the National Railway Masterplan (RIPNAS) 2018 (DJKA, 2018). Specifically, Freight OD patterns for Java as depicted in RIPNAS 2018 indicates relatively low volume of transport demand between DKI Jakarta, West Java and East Java (Fig. 6). This is rather strange considering these provinces are the top three provinces with the highest GDRP and population on Java. In addition, the railways connecting these provinces along the Northern coastline of Java have the highest capacity on the island, supported by terminals with most adequate container handling equipment such as reach stackers. On the contrary, the OD patterns estimated using gravity model indicate a substantial transport demand volume from DKI Jakarta, West Java, and Surabaya and vice versa. This difference has led to a higher value of rail tonne-kilometers predicted by my model compared to the reported statistics. In terms of average distance travelled, my model output shows a slightly higher average rail transport distance (498 km) compared to the available data (450 km). This difference could also stem from the differences between available and estimated OD patterns.

Table 2
Validation of model estimates with available statistics.

Variable	Available statistics	Model estimate
Java's rail freight volume 2015	7.69 Million Tonne	7.63 Million Tonne
Java's rail freight Tonne-Kilometers 2015	3490 Million Tonne-Kilometer	5410 Million Tonne-Kilometer
Average distance travelled	450 KM	498 KM

Unfortunately, reliable OD data in terms of volume and tonne-kilometers for all modes of transport is currently unavailable. The available 2011 OD data from the ministry of transport records unrealistically high transport volumes between districts on Java with a total volume of >11 billion tonnes as reported in de Baat et al. (2015). This underlines the need to establish a reliable national OD data broken down by modes that can be used to calibrate available and future transport models.

The result of this calibration and the problems highligted with the lack of reliable OD data imply that the model might suffer from inacuracy when used to predict the impact of modal shift measures on specific routes, particularly those between OD pairs with freight volume that differs significantly than the output of the gravity model. Furthermore, higher rail tonne-kilometers will also lead to higher estimate for carbon emissions from rail compared to emissions which are calculated based on the reported statistics. Nevertheless, due to small modal share of rail and relatively lower rail's emission factor this discrepancy is expected to have a marginal effect on the total CO_2 estimates. Considering the abovementioned conditions, the model is best suited to predict the aggregate impact of policy measures on the aggregate emissions and less suited for predicting the impact of measures at a detailed geographical resolution such as at route level.

3.4. Route choice model

Route choice model is a part of transport modelling technique typically deployed to analyze the impact of transport policy measures on the route choice of shippers, which shapes the spatial pattern of freight traffic on a transport network.

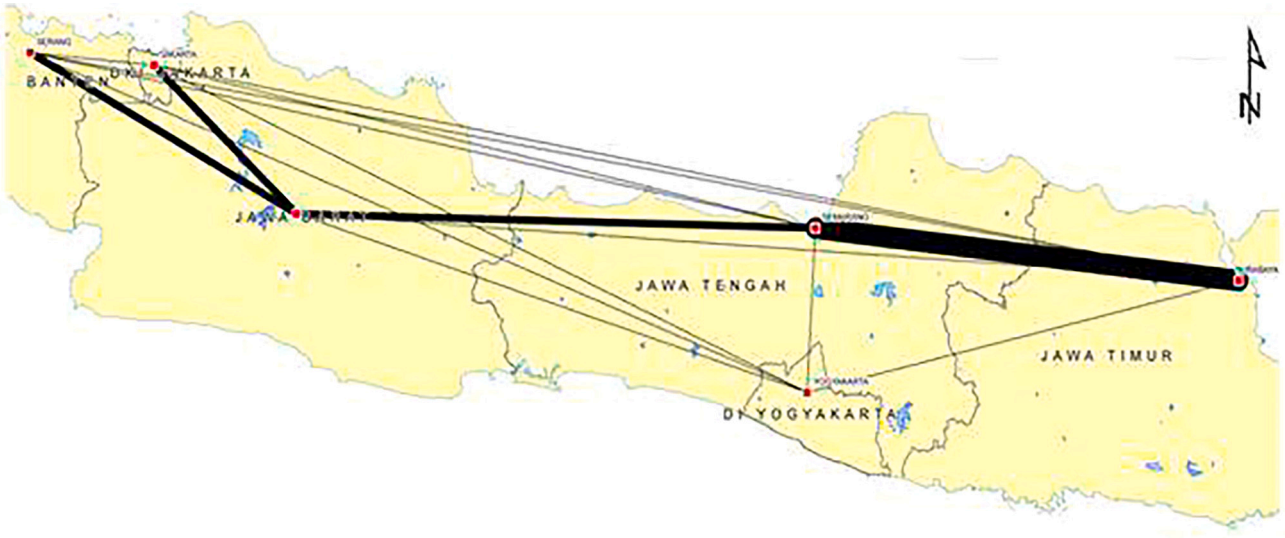


Fig. 6. Projection of freight transport demand pattern on Java for the year 2030 based on OD data for the year 2011 (Source: RIPNAS 2018 (DJKA, 2018)).

A path size logit model is used in combination with a path generation method to assign the volume of freight transport across all possible land routes between all origins and destinations on Java. The shortest path algorithm and a choice set generation heuristics are used to identify the routes used by freight forwarders. The model accounts for rail connections as well as the first and last mile road connections between the origin and rail terminals and between rail terminals to end destination. The path size logit model takes overlaps between the alternative routes into account and distinguishes the transport costs associated with these alternatives properly. The basis of this model can be found in (Ben-Akiva & Bierlaire, 1999). The route probabilities are given by:

$$P_r = \frac{e^{-\mu\,(C_r + \ln S_r)}}{\sum_{h=1}^{H} e^{-\mu\,(C_h + \ln S_h)}} \tag{5}$$

while the path size overlap variable S is defined as follow:

$$S_r = \sum_{a \in LK_r} \frac{Z_a}{Z_r} \frac{1}{N_{ah}} \tag{6}$$

where:

P_r = the choice probability of route r,
C_r = generalized costs of route r,
C_h = generalized costs of route h within the choice set,
CS = the choice set with multiple routes,
h = path indicator/index, $h \in CS$,
μ = logit scale parameter,
a = link in route r,
S_r = degree of path overlap,
Lk_r = set of links in route r,
Z_a = length of link a,
Z_r = length of route r,
N_{ah} = number of times link a is found in alternative routes.

3.5. Emission estimation

The Activity, Structure, Intensity, Fuel (ASIF) framework is used to assess the impact of different ambition levels of the action program on the total CO_2 emissions of freight transport on Java. The output of JFM provides the tonne-kilometer for different modes and routes ("Activity and structure"). Four vehicle types are considered in the model: rail, heavy truck, medium truck, and light truck due to the differentiation in fuel intensity (liter fuel consumed/km) of each truck type. Based on the aggregate tonne-kilometer for truck and rail mode produced by JFM and the average load factor data of each vehicle type, the vehicle kilometer of each vessel type is estimated ("Structure"). I use the data from the Motor Vehicle Weighing Implementation Unit (UPPKB) to determine the average load factor for each vehicle type. Furthermore, the fuel consumptions of all vehicle types are computed using fuel intensity across different vehicle types ("Intensity"). The resulting fuel consumption for each vehicle type is then used to estimate the total CO_2 emissions using carbon factor and energy content data for different fuel types ("Emission factor"). This study focuses on direct tank-to-wheel emission from road and rail transport (scope-1).

To calculate the emission reduction potential, the emission under baseline scenario (where no mitigation measures are implemented) is compared to scenarios where mitigation actions are implemented. The following equation describes the way emission is calculated.

$$E_{mf} = A_m . S_{mf} . I_{mf} . F_f \tag{7}$$

$$TCO_2 = \sum_{m=1}^{M} \sum_{f=1}^{F} E_{mf} \tag{8}$$

where:

TCO_2 = total CO_2 emissions from all modes (road and rail transport,)
E_{mf} = total CO_2 emissions from transport mode m using fuel type f (in tonne CO_2),
A_m = total annual activity for mode m (in tonne kilometer),
S_{mf} = share of mode transport m with fuel type f (in percentage).
I_{mf} = engine intensity of *mode m*, fuel type f (in Liter fuel/tkm),
F_f = emission factor of fuel type f (in tonne CO_2/Liter fuel).

3.6. Data

Data used as input for the modelling exercise is obtained from various initiatives which include online stated preference surveys, bilateral stakeholder interviews, and estimation based on available secondary data. The quality and availability of data used in a modelling exercise determines the depth and reliability of the analysis. As concluded in Section 2.7 there is a huge gap both in data quality and availability that can be used to perform comprehensive quantitative analysis. Hence, in this study, a significant effort has been devoted to gather and estimate the missing data. The detailed description for each input data is as follows:

Table 3
Choice set for stated preference survey.

Choice set	Transport cost (IDR)	Lead time (h)	Risk and safety (%)	On-time reliability (%)	Choice
Road	13,000,000	24	95	90	
Rail	10,000,000	16	85	70	
Sea	5,750,000	120	90	80	

- **Data for origin and destination or transport demand volume**

This data is obtained from the research and development division of MoT (Puslitbang). I used 2011 OD data as a basis to estimate transport volume between origins and destinations in the year 2011 and to project the transport demand until 2030. Despite having the sufficient origin and destination coverage, the data records unrealistically high transport volumes between districts on Java as reported in (de Baat et al., 2015). As a result, an estimation procedure as described in Section 3.1 was deployed to estimate more realistic transport volumes between districts, taking into account all relevant socio-economic variables.

- Data for unit **transport costs** (IDR/t kilometer)

This data is obtained from interview with leading freight forwarders and logistics service providers. Furthermore, the transport cost for all routes is estimated using the shortest path algorithm applied to the network model.

- Data for **mode choice** model

To obtain this data, two stated preference surveys were conducted in the period of February – July 2020. The first survey was done in-person through a workshop carried out by GIZ in February 2020 in Jakarta and the second survey was conducted online through a web-based interface in April 2020. The respondents of the survey were from diverse stakeholder groups which are invited to the workshop by the Indonesian ministry of transport and to be partner of the project. They represent are cargo owners, shippers, unimodal and multimodal freight forwarders, logistic service providers, and shipping lines who are knowledgeable about road and rail modes. The respondents also represent shippers of various commodities namely food and beverages, cement, fuel oil, chemicals, coals, electronics, clothing, and textiles, and electronics. The majority of the respondents (68%) are shippers of containerized consumer goods such as food and beverages, clothing, and electronics. In terms of firm size, most of the participants (76%) represent large companies while the rest (24%) belong to the category of small company.

Each survey consists of nine choice observations for transporting 1 TEU or a full cargo load of a heavy-duty truck (on average 15 t) between Jakarta- Surabaya. This route is chosen as it represents a long-haul freight shipment which has the most volume and potential[3] for rail to offer competitive advantage against road freight. Contextualization of the choice scenarios is specified by the type of commodity, travel distance, value and volume of the cargo, and cargo containerization. Each choice set presents three different unlabeled alternative modes (i.e. road, rail, sea), four attributes, and three levels. Table 3 provides an example of the choice set. The design of the choice sets follows an optimal orthogonal design where each attribute level is presented in an independent manner with other attributes and with the same amount of time and over the choice sets (Street, Burgess, & Louviere, 2005). This design also helps avoid correlation between attributes of the modes.

In total twenty-seven respondents provided their responses, resulting in 243 choice observations. Nevertheless, responses of 7 respondents were dropped due to observable pattern of invalid answers. For instance, some respondents did not finish the survey or provided the same answer for all the choice sets. After the invalid responses are filtered out, a total of 180 choice observations from 20 respondents are obtained. Based on the rule of thumb for ideal sample sizes suggested by Orme (1998), I acknowledge the need to have a higher number of respondents to help produce more robust model parameters. Nevertheless, the estimated parameters could contribute as an initial insight into the shippers' preferences in modal choices and their potential substitutions for the purposes of the study. A follow up study will be needed to gather higher number of respondents which also cover commodity types that were underrepresented such as palm oil, fuel, coal, and chemical products. It is noteworthy that the low response rate could be attributed to disruptions caused by COVID-19 pandemic which caused stakeholders to be less interested to participate in the survey. In addition, the research project also faced a limitation in budget and time that made it difficult to expand the timeline and tasks of the project.

The description for each of the attributes is as follow:

- **Transport cost:** The total costs for shipping the goods from origin to final destination with either one or multiple modes of transport. Transport costs is the main attribute of the modes.
- **Lead time:** Door-to-door travel time of 1 TEU from origin to final destination.
- **Risk and safety:** The likelihood of no loss, damage, or threats on cargo, driver and transport mode during a shipment process. For instance, 95% risk and safety means that 95% of the shipments and the driver arrive safely at the destination without loss/damage.
- **On time reliability:** The percentage of on-time delivery for shipping goods. Higher on-time reliability means higher percentage that goods are shipped on schedule from origin to destination. For instance, 95% on time reliability means that 95% of the shipment arrives at a scheduled/expected time.
- **Data for emission model**

Data for computing the total CO_2 emissions of freight transport on Java is obtained from ICCT's estimate of **engine fuel intensity** (I) for rail and different types of trucks (see Table 4): light, medium, and heavy. Data for estimating the **average load factors** of the available modes are obtained from PT.KAI and the Motor Vehicle Weighing Implementation Unit (UPPKB) in Losarang and Balonggandu. Data for emission factor of diesel fuel is obtained from PPTMBG-Lemigas study, where the EF of the national diesel fuel is 2.6 kg of CO_2/l (Novie, 2017). Emission factor for other fuel types such as biodiesel is estimated based on the methodology explained in (Coronado, de Carvalho, & Silveira, 2009). Specifically, a carbon reduction factor is applied to estimate the carbon content of diesel fuel as shown in Table 5.

Table 4
Engine intensity for different vehicle types.

Type of trucks	Engine intensity (Liter diesel/km)	Source
Light commercial trucks	0.08	ICCT estimates
Medium freight trucks	0.25	ICCT estimates
Heavy freight trucks	0.46	ICCT estimates
Rail	4.4	PT.KAI (PT.KAI, 2020)

- **Data for transport network schematization**

Geographical Information System (GIS) data is used to schematize the transport networks for all relevant transport modes on Java: road, rail, and sea. The schematization for road network is obtained from an openly available map (https://www.diva-gis.org/gdata), data for rail

[3] Based on a survey and stakeholder consultation conduceted by GIZ for freight forwarders, logistics service providers, and shippers in technical design study (GIZ, 2021).

Table 5
Emission factor for diesel fuel and biodiesel.

Indicator	
Amount of biodiesel (%)	10
Net reduction in CO_2 (%)	8
Emission factor of biodiesel (gCO_2/l)	2638
Original emission factor of diesel fuel (gCO_2/l)	2660

tracks is obtained from the DG Railways, data for rail stations are obtained from a commercial provider (www.lapakgis.com) and data for sea network is schematized using QGIS software. There are 459 rail stations included in the schematization, which represent active rail network that serve freight transport for 118 centroids/cities across Java. Fig. 7 provides the schematization of Javanese multimodal transport networks.

4. Policy scenarios

In this section I present the description of policy measures, their combination, and stringency levels. Which form scenarios that are tested using my modelling framework. These measures are applicable to all segments of intermodal rail transport on Java, particularly on the Java's northern coastline (Pantura) and Southern coastline corridor. The focus of demand scenario development is to take into account uncertain developments and factors that can influence the magnitude of freight transport on Java., modal share and transport emissions for the period between 2011 and 2030.

4.1. Baseline scenario

In the baseline trade scenario, I assume that Java island's GDRP (at constant 2010 prices) grows at an average annual rate of 4.78%[4] until 2030. This is equivalent to GDP growth roughly from Rp. 4200 trillion in 2011 to Rp. 10,000 trillion by 2030. Here I take into account the impact of the economic crisis due to COVID-19 pandemic by applying a low Compounded Annual Growth Rate (CAGR) of Java's GDRP in 2020 which is at 0.5% as projected by IMF in their October 2020 projection (Akhlas, 2020)-a major reduction from 5.51% in the year 2015 until before 2020 (Fig. 8).

This low growth rate has caused stagnation in domestic trade volume between 2020 and 2022. However, I also assume that the economy will bounce back in 2022, which translates into an annual growth of 7.1% by 2025. Furthermore, the scenario envisions that the CAGR's of Java's GDRP is going to return to the 6% rate –similar to that prior to the crisis in 2020. Fig. 9 provides the projection of the trade volume on Java based on these assumptions. Table 6 provides an overview of the key assumptions used to project freight volume on Java until 2030. A more detailed description of the regression model used to project freight transport volume can be found in Annex 1.

4.2. Policy ambition scenarios

This study analyses the expected impact of a basic, moderate, and strong ambition scenario compared to a baseline scenario. These scenarios consist of eight interventions from the intermodal action program with different ambition levels for each scenario. Table 7 provides an overview of the assumed changes in the variables of the transport system that are tested in the modelling exercise.

[4] The average CAGR of 4.78% is computed based on historical data on Java's economic growth and projection from IMF until 2021.The underlying method used to carry out the forecast is based on the work in Indriana (2019). *Adaptive policy decision pathways for robust shipping network investment planning in Indonesia* Technical University Delft]. Delft.

- **Basic**

This scenario describes a minimum level of intervention measures that are focused on tackling primary problems that have caused intermodal rail transport to be less competitive than road transport. Compared to the other 2 ambition levels, the measures pose the minimum political or financial barriers and hence are designed to be implementable within short-term. In general, the basic intervention focuses on reducing additional costs, time, and risk aspects of intermodal rail transport so that it can be on a level competition with road transport.

In terms of costs, the measure includes reduction in container transloading (lift-off/lift-on) costs by 30% at each intermodal station. Furthermore, the government could provide a tax benefit for freight forwarders that use rail freight transport. In this scenario I assume a small tax reduction, which could translate to a 5% cost reduction for the first and last mile transport (i.e. truck trips from and to intermodal rail stations). In addition, I also test the impact of reducing Track Access Charges (TAC) to by 10% of the total rail transport costs.

In terms of time, the scenario envisions a possible improvement in cargo transloading time and waiting time at the terminal as a part of initiative to improve service quality, and the reliability of train scheduling process. In this scenario both waiting time and cargo transloading time are reduced by 25%. Additionally, this scenario also reflects improvement in the quality and capacity of road network that provides access to the existing dry ports on Java. This is expected to reduce travel time to access the dry ports by 30%.

In terms of risk and safety, I test the impact of increasing customer's confidence on the safety and risk of using intermodal rail transport through a higher commitment to prevent damage and loss throughout the handling and transport process. Hence, I assume a 10% increase in safety and risk rating to 0.85.

- **Moderate**

Moderate measure represents a higher degree of commitment in reforming the intermodal transport sector. In this scenario, stronger financial aid, investments, and service quality management are deployed to deliver improvements in cost, time, and risk and safety of intermodal transport. Specifically, I assume that there will be further reduction in transloading cost (−40%), TAC (−20%), first mile and last mile costs (−7.5%), total transloading time at the rail terminals (−40%), and an increase in risk and safety rating to 0.95.

This scenario reflects the use of additional handling equipment such as reach stackers, or gantry cranes to further increase the efficiency of cargo handling process such that the costs and the time to transload the cargo from trucks to rail can be reduced. Furthermore, apart from all the improvements that are covered in basic intervention, this ambition level also takes into account the development of rail connection to Jakarta International Container Terminal (JICT) and an elimination of Value Added Tax (VAT) which amounts to 10% of the total rail transport cost.

- **Strong**

The strong intervention scenario includes all measures that are specified in the basic and moderate measures with higher level of progression. Specifically, this scenario considers reduction in transloading cost (−50%), TAC (−30%), first mile and last mile costs (−10%), total transloading time at the rail terminals (−50%), and increase in risk and safety rating to 0.99. The distinct intervention scale in this scenario, is in the development of rail connection to the new freight terminal Teluk Lamong in Surabaya which is developed as a new alternative port to the congested Tanjung Perak Port in Surabaya. Additionally, it also features a risk and safety rating that is improved to a high level (99%) – higher than that of road transport at 95%. This is reflecting the strong commitment in the maintaining high service quality for intermodal rail transport such that the risk of cargo damage and loss is minimized.



Fig. 7. Multimodal freight transport network on Java source: own schematization.

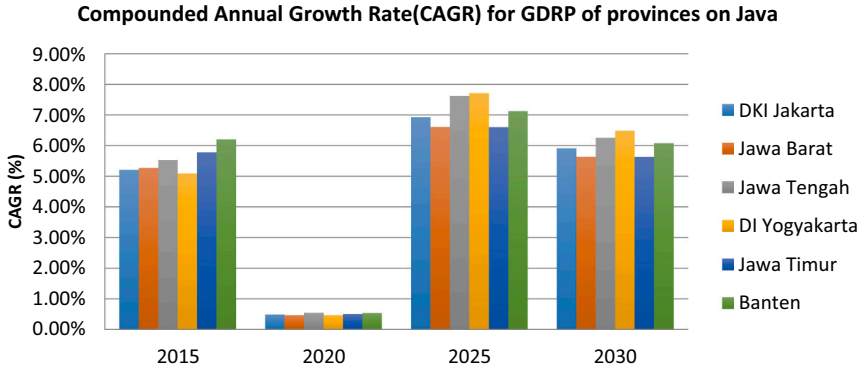


Fig. 8. Projection for Java's CAGR until 2030.

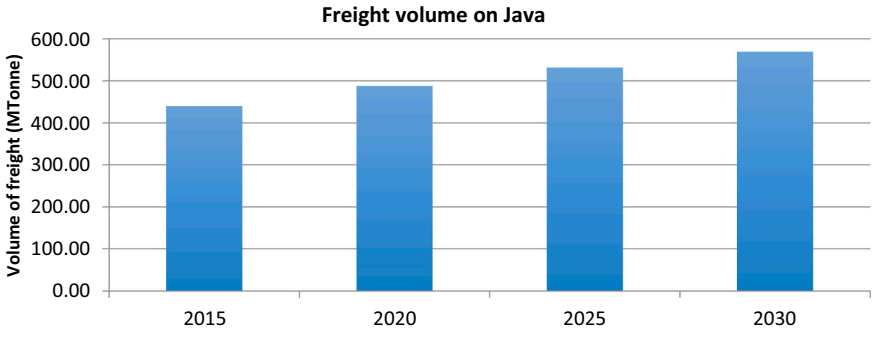


Fig. 9. Freight transport volume on Java until 2030.

Table 6
Assumption for Java's economic growth scenario.

Economic Indicator	Assumption
Java's GDRP Growth 2020	4.78%
Java's CAGR 2020	0.5%
Java's GDRP Growth 2025	7.71%
Java's CAGR 2020	6%

5. Impact of policy intervention scenarios on CO_2 emissions

5.1. Baseline scenario

The baseline scenario projects land freight emissions in which no policy interventions to increase the modal share of rail – "the intermodal rail action program" are implemented. However, I included the impact of current government's policies which mandates the incorporation of biofuel into diesel fuel – also called biosolar from 2010 until 2030. I incorporated these policies in my modelling based on the timeline of the

Table 7
Action programme with three different ambition levels.

No	Intervention measure	Baseline	Basic	Moderate	Strong
Trade					
	No trade policy intervention	4.78% with CAGR of 0.5% in 2020	Baseline assumption	Baseline assumption	Baseline assumption
Costs					
1	Reducing transloading costs	Transloading cost is 550,000/lift	−30% (IDR 165,000/TEU)	−40% (IDR 220,000/TEU)	−50% (IDR 275,000/TEU)
2	Tax benefit for trucking companies that serve intermodal trips	No improvements	−5%	−7.5%	−10%
3	Eliminating VAT	VAT is 10% of transport costs	No reduction of VAT	−100%	−100%
4	Reducing TAC	TAC is 55% of the total rail transport costs	−10%	−20%	−30%
Time					
5	Improving existing road connections to rail stations and dry ports	No improvements	Reduction in travel time to dry ports by 30%	- Reduction in travel time to dry ports by 30% - Connecting JICT to rail network: −20% in rail total travel time from and to JICT	- Reduction in travel time to dry ports by 30%. - Connecting JICT and Teluk Lamong to rail network: reduced rail total travel time: from and to JICT (−20%) from and to Teluk Lamong (−30%)
6	Reducing transloading time in terminal	1 h/TEU	0.75 h	0.6 h	0.5 h
7	Reducing waiting time in terminal	1 h/TEU(Truck)	0.75 h	0.6 h	0.5 h
Service quality					
8	Improving service quality of rail transport in terms of risk and safety	75% of cargoes arrive on time, without damage and loss	10% improvement	20% improvement	24% improvement

implementation of the measures to estimate the total GHG emissions of freight transport on Java. In my scenario, the amount biofuel incorporated in the solar fuel is gradually increased from 10% in 2010, to 15% in 2015, and to 30% in 2020. I assume that this fuel composition will remain the same until 2030.

Based on my modelling, I estimate that in 2020, the land transport modal split on Java (by weight) is 98.26% for road and 1.74% for intermodal rail transport. Note that this estimate differs than estimate in other studies such as RITN 2019. This could be caused by a number of factors, but it is expected that the major difference may come from the data source used. In the absence of reliable freight transport volume for Java or OD data, this study built on the estimation of Java's freight volume in (de Baat et al., 2015) which uses a gravity modelling approach which is cross-validated with the 2011 OD data. The harmonization of different OD data sources is an exercise that I envision to be valuable when new data sources become available. In this study, I use my own estimate of rail and road modal share as the baseline scenario with which the impacts of policy measures will be evaluated and compared.

My projection shows that CO_2 emissions from rail and road freight are to increase up to 17.34 MTonne until 2030 (see Fig. 10) where it is 50.6% higher compared to the 2015 level (11.5 MTonne) and 38.6% higher than 2020 level (12.5 MTonne).

The growth of emissions between 2015 and 2030 periods are affected by three primary factors:

1. The growth in the number of vehicles and vehicle-kilometers due to the enforcement of zero ODOL (over dimensioned and over loading) trucks policy[5] until 2030. Gradual reduction of the number of overloaded trucks will translate to higher total vehicle-kilometers for road mode due to the significant reduction of load factor for heavy and medium trucks. Currently, it is estimated that around 50% of medium trucks[6] in Indonesia are overloaded. The excess in payload, in many cases, could reach 100% higher than the officially allowed loading capacity.[7] Table 8 provides the average load factor of different modes used in estimating CO_2 emissions

2. The use of biosolar with lower carbon content (up to 30% lower) compared to diesel fuel as mandated in government regulation. The increase in the mix of biofuel is laid out in the Table 9

[5] ODOL policy is established by Directorate General of Road Transport (DJPD) to minimize the number of trucks which dimensions and total weight exceed the legal limit. Under this policy, trucks that are overloaded and over-dimensioned are given substantial fine to discourage violations against the law.

[6] Based on the information retrieved from https://otomotif.kompas.com/read/2020/12/16/070200715/truk-odol-bikin-bengkak-biaya-pemeliharaan-jalan-tol-jakarta-cikampek

[7] Based on the data retrieved from weighing-in-motion (UPPKB) Balonggandu and Losarang. For medium trucks with average gross vehicle weight between 5 and 10 t, the average permissible payload is on average 4.3 t/vehicle. For heavy trucks with average gross vehicle weight > 10 t, the average permissible payload is 13.3 t. Based on the observed weights registered at the weighing-in-motion station, overloaded medium trucks on average carry 8.6 t cargo while overloaded heavy trucks on average carry 20 t of payload.

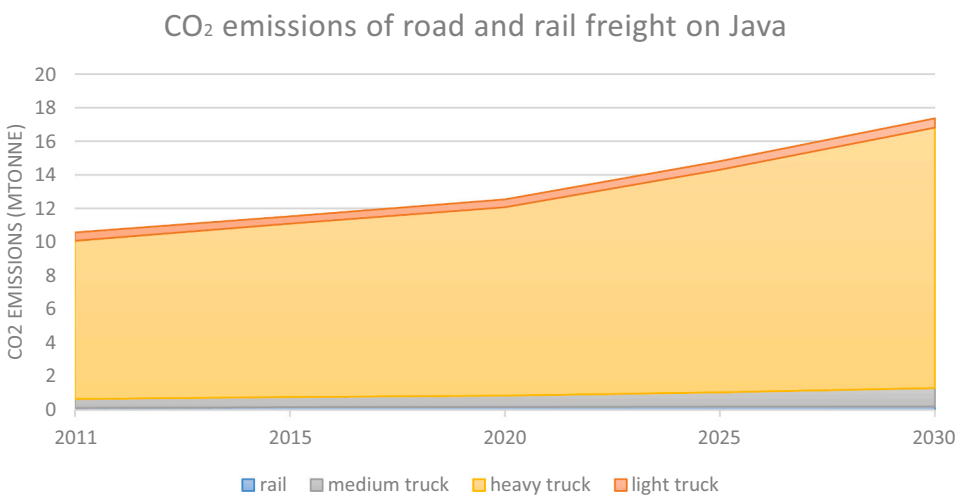


Fig. 10. CO_2 emissions for Baseline scenario.

Table 8
Average load factor between 2011 and 2030 taking into account overloaded and over dimensioned trucks.

Mode	2011	2015	2020	2025	2030
Rail (tonne/vehicle)	420.00				
Road light (tonne/vehicle)	0.82				
Road medium (tonne/vehicle)	6.45	6.88	6.45	5.59	4.73
Road heavy(tonne/vehicle)	16.65	17.32	16.65	15.31	13.97

Table 9
Assumption on the emission factor of fuel types between 2010 and 2030.

Indicator	2010	2015	2020	2025	2030
Amount of biodiesel (%)	10	15	30	35	40
Net reduction in CO_2 (%)	8	12	24	28	32
Emission factor of biodiesel (gCO_2/l)	2638	2612	2468	2399	2319
Original emission factor of diesel fuel (gCO_2/l)	2660	2660	2660	2660	2660

3. The overall growth of transport demand between 2020 and 2030, taking into account lower economic growth in 2020–2021 due to the economic slowdown

5.2. Impact of the intermodal action programme on modal share and CO_2 emissions

I assume that the intermodal action program will be implemented gradually from 2022 onwards and therefore the impact of this program on the modal share of rail will also be realized accordingly. Fig. 11 presents the combined impacts of the policies and measures in each scenario on the tonne-kilometers of road and rail freight. Compared to the baseline, a basic level of ambition increases rail modal share from 1.74% to 7.12% by 2030, while medium and strong ambition increase rail share to 14.87% and 19.59% respectively. This increase in rail modal share results in a strong increase in rail's Tonne-kilometers and a reduction in road Tonne-kilometers for each ambition level (Fig. 11). Specifically, moderate ambition will result in rail's tonne-kilometers seeing 9 times increase, while strong ambition will result in 11.3 times compared to its 2020 level. Furthermore, the increase in rail tonne-kilometer also increases rail modal share in tonne-kilometers quite significantly namely to 13.93% under basic ambition, to 27.8% under moderate ambition, and up to 35.23% under strong ambition (Fig. 12).

In terms of tonnage, the market for rail transport is projected to increase by 30.7 MTonne (Basic ambition), 74.8 MTonne (Moderate ambition), and 101.7 MTonne (Strong ambition) compared to the baseline. Table 10 provides an overview of rail freight volumes across different intervention measures. Fig. 13 visualizes the rail freight volume under baseline and strong ambition scenario in the year 2030. It can be observed that the southern corridor could generate high demand for rail transport.

In terms of GHG mitigation potential, increased rail modal share may reduce CO_2 emissions considerably with emission reduction ranging from 1.26 to 3.74 MTonne in 2030 compared to baseline (Fig. 14). Specifically, CO_2 emissions in the year 2030 may be reduced by 1.26 MTonne (basic ambition), 2.89 MTonne (moderate ambition), and 3.74 MTonne (strong ambition).

In relative terms, basic ambition could reduce CO_2 emissions by 7.3% in 2030, while moderate and strong ambition could deliver 16.7% and 21.6% reduction respectively by 2030 Fig. 15.

It is worth highlighting that the cumulative CO_2 mitigation that can be attained due to the modal shift to rail freight between 2022 and 2030, is considerably high. Table 11 provides a summary for the CO_2 mitigation potential of each intervention measure. While basic intervention measure reduces around 8.36 MTonne CO_2, the strong measure is expected to mitigate around 24.72 MTonne CO_2.

Aside from the environmental benefits of CO_2 reduction, there are also significant economic benefits, which are aligned with government's development objectives, namely improvement in connectivity and reduction of logistics costs. Lower transport costs may, in the long run, lead to growth in trade volume, employment, and GDP, lower amount of subsidy needed for diesel fuel as a result of modal shift and potentially increased income from tax from the intermodal transport sector. I calculated the total transport costs to the society. The total transport costs are computed by multiplying transport cost of each route (IDR/Tonne-kilometer), the volume of goods transported using road and rail on each route (Tonne-kilometer), and the value of time for both road and rail transport to serve transport demand on Java by 2030. Table 12 provides a summary of the economic impacts. It is estimated that basic intervention would result in a reduction of 2318 Billion IDR of transport costs (1.40%), while moderate and strong measures would result in reduction of 7680 Billion IDR (4.6%) and 12,543 Billion IDR(7.6%) respectively. These potential reductions are quite high considering the amount of savings that can be attained by society.

6. Policy implications

This section discusses policy implications based on previous section. They are as follows:

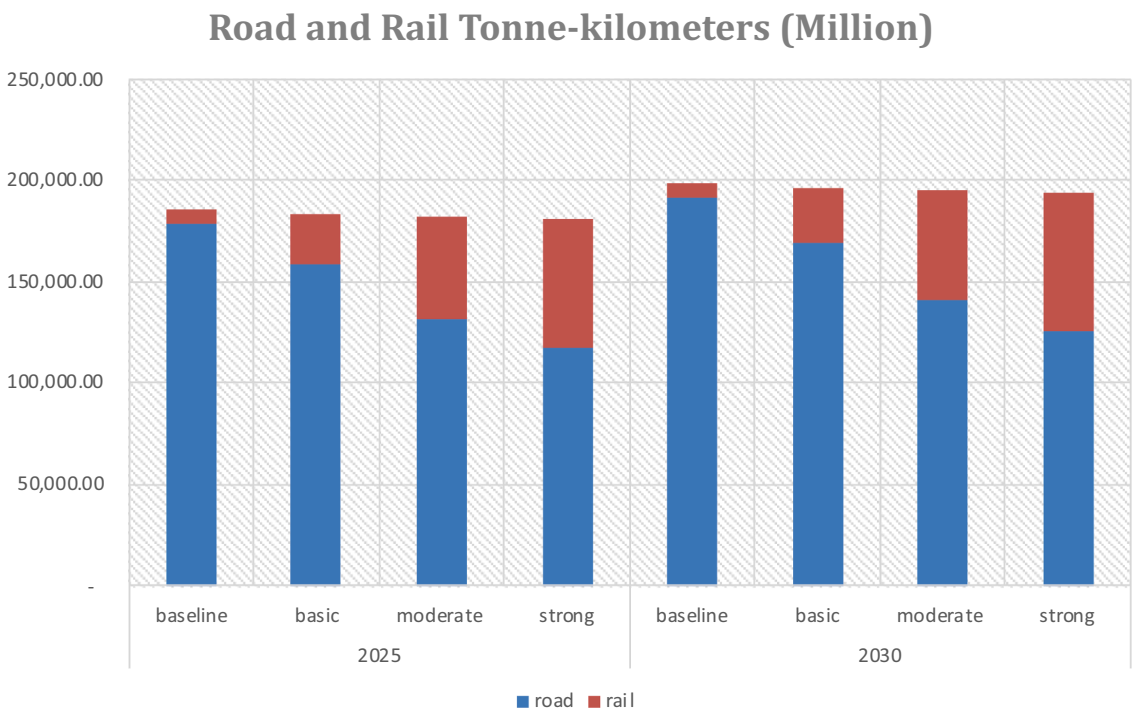


Fig. 11. Tonne-kilometers of rail and road under different ambition levels.

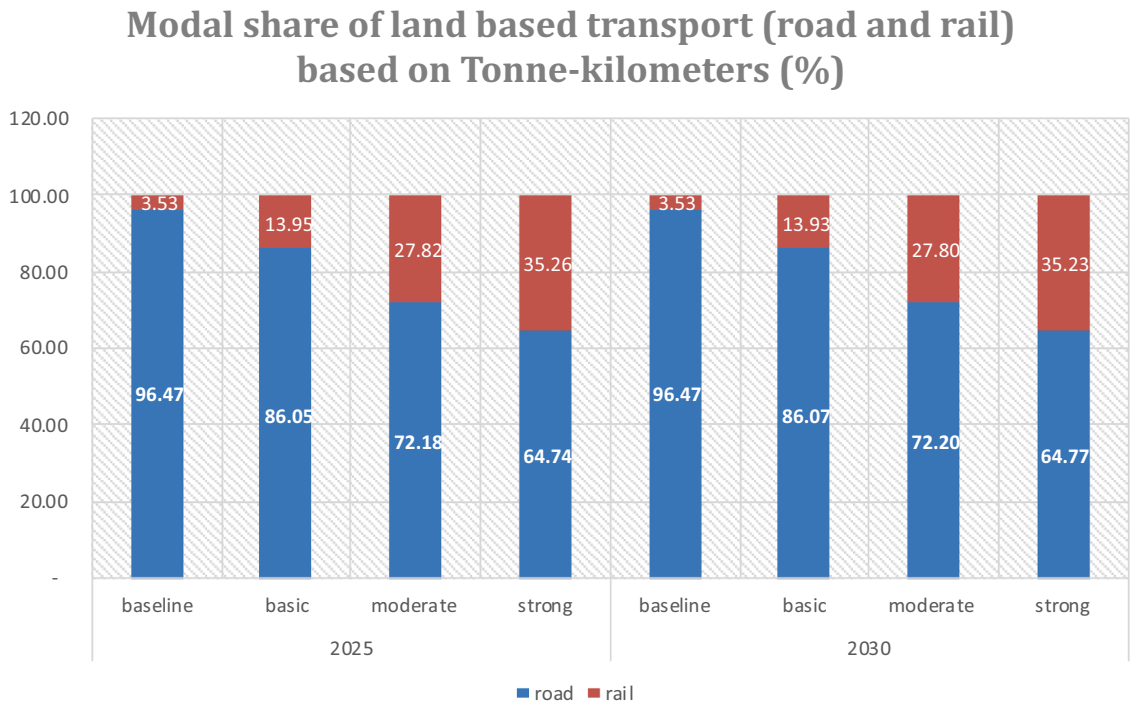


Fig. 12. Modal share of road and rail based on tonne-kilometers.

Table 10
Impact of intervention measures on rail freight volume on Java.

Policy intervention scenario	2020	2025	2030	2020–2030 growth (%)
Baseline (MTonne)	8.49	9.23	9.86	
Basic (MTonne)	8.49	38.03	40.56	378
Moderate (MTonne)	8.49	79.37	84.69	898
Strong (MTonne)	8.49	104.52	111.53	1214

1. Carrying out a comprehensive cost-benefit analysis

The results of my assessment have provided insights into the effectiveness and benefits of modal shift policy measures in reducing CO_2 emissions from freight transport as well as transport costs. Nevertheless, the costs of implementing these measures across different ambition levels have not been studied thoroughly. While every tonne of cargo shifted from road to rail will lead to a net reduction of these

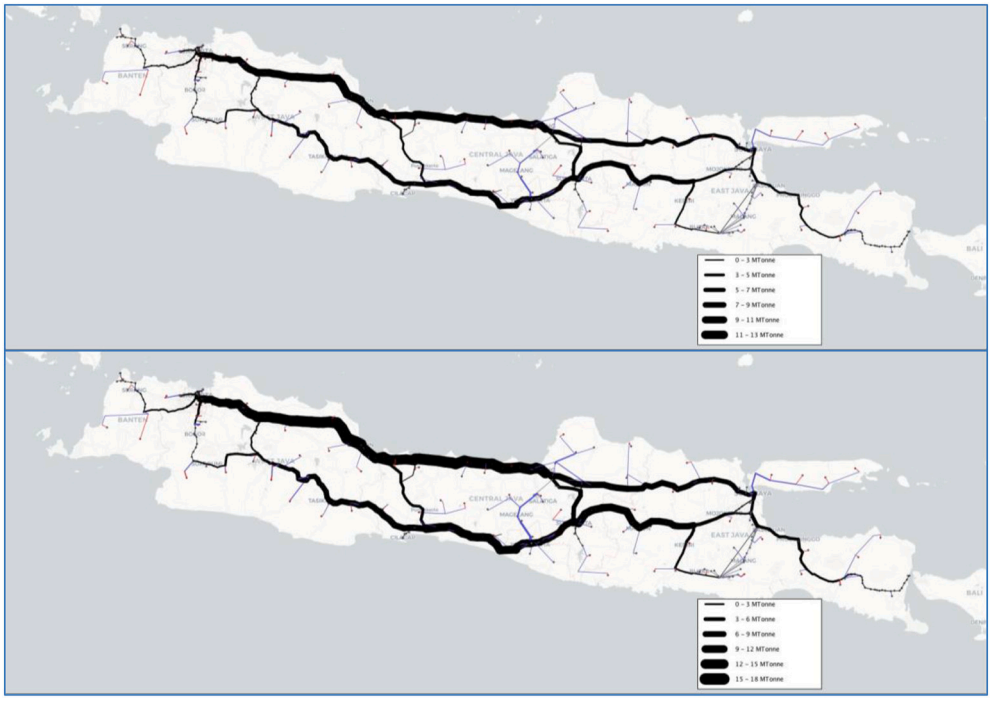


Fig. 13. Rail freight transport pattern on Java by 2030 under baseline scenario (top) and strong intervention scenario (bottom).

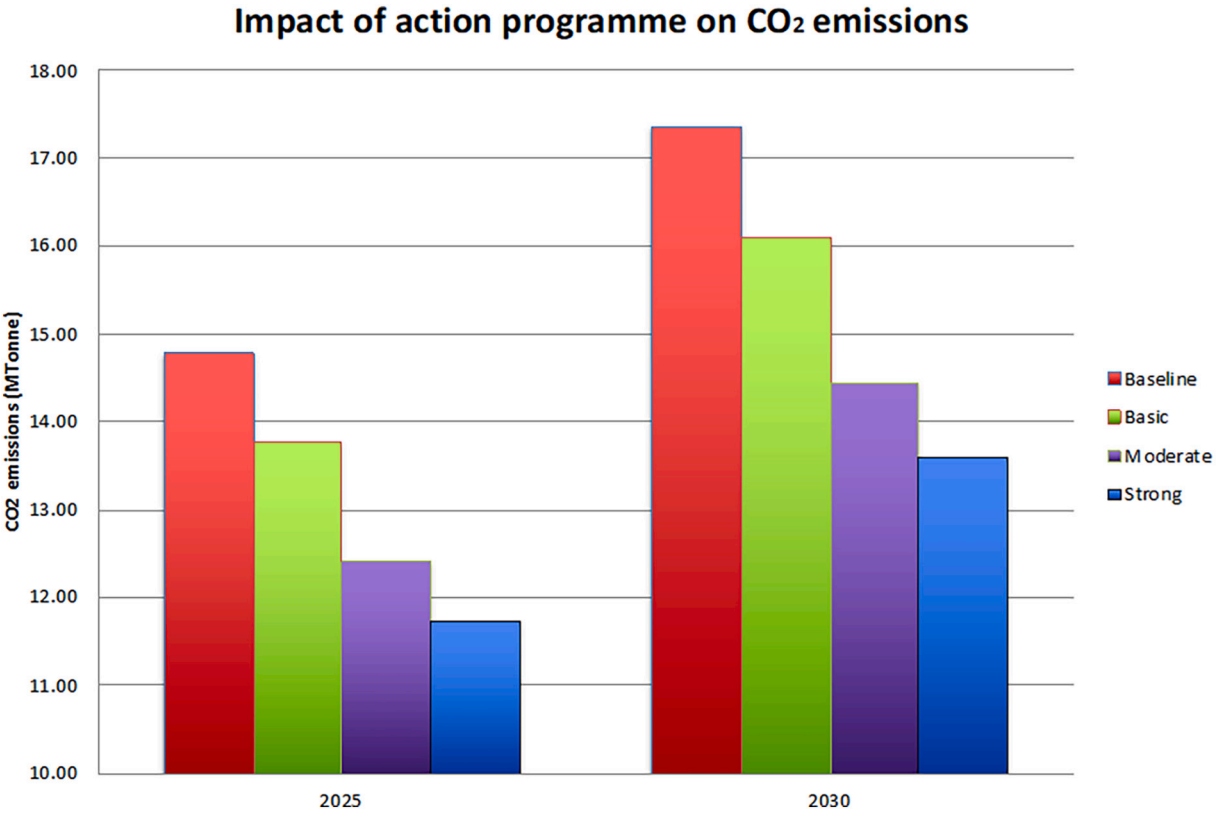


Fig. 14. Impact of intervention measures on CO_2 emissions from freight transport on Java.

externalities, a proper social cost-benefit assessment of the investment cannot be done without good insight in actual cost levels. Hence, a financial analysis is valuable to estimate the costs of investment needed to develop adequate intermodal infrastructure, subsidy needed, and loss of government revenues to implement these measures. By contrasting these costs and CO_2 mitigation attainable, the cost-effectiveness of modal shift measure can be determined in terms of IDR/TCO_2 reduced and compared with other promising alternative measures e.g. the use of low-carbon alternative fuels such as biofuels, or the establishment of fuel efficiency and emission standard for road freight. A marginal abatement

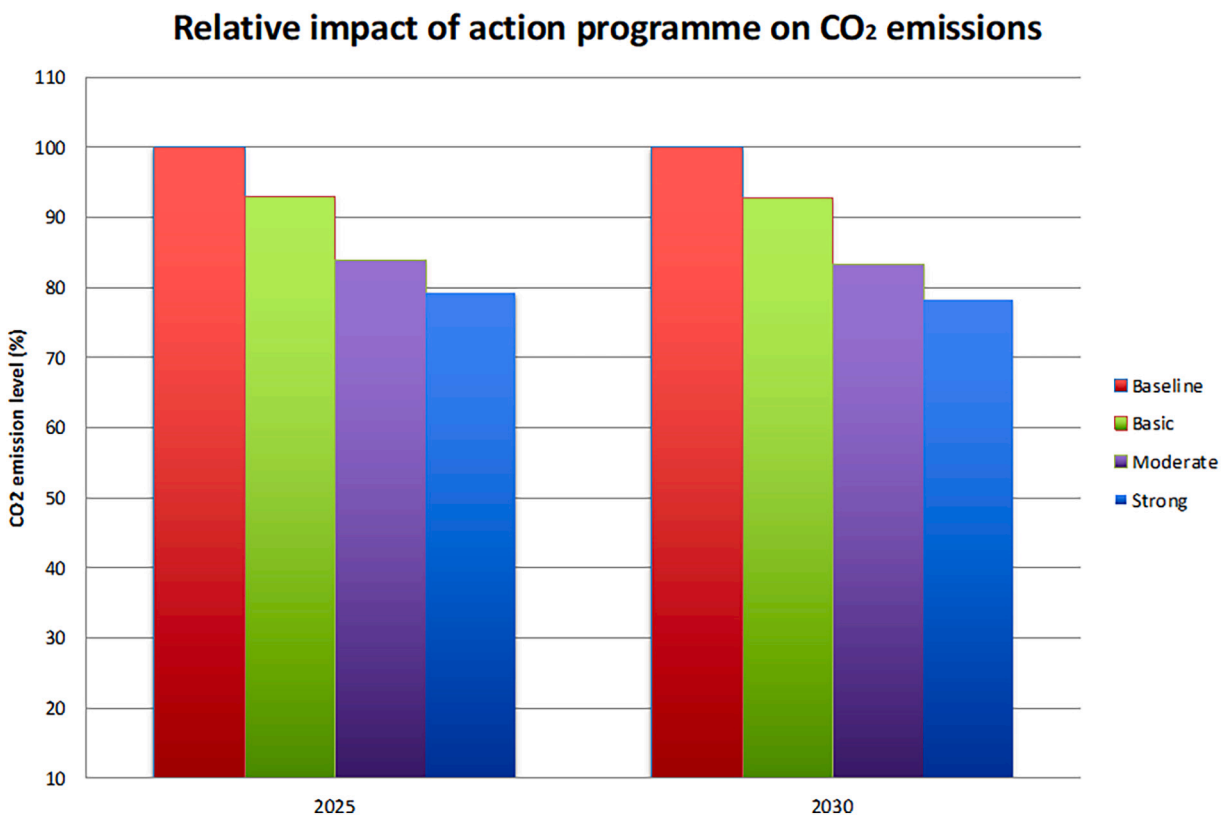


Fig. 15. Relative impact of action programme on CO_2 emissions.

Table 11
CO_2 reduction due to modal shift to rail transport on Java.

Policy intervention scenario	Land freight transport emissions (MtCO2)			Emission reduction (MtCO2)	
	2020	2025	2030	Annual reduction in 2030	Cumulative
Baseline	12.51	14.79	17.34		
Basic	12.51	13.75	16.08	1.26	8.36
Moderate	12.51	12.42	14.45	2.89	19.11
Strong	12.51	11.72	13.6	3.74	24.72

Table 12
Transport costs reduction due to modal shift to rail transport on Java.

Intervention policy	Total transport cost paid by shippers (Billion IDR)	Reduction in total transport cost (Billion IDR)	Reduction (%)
Baseline	165,442.51		
Basic	163,124.53	2317.97	1.40
Moderate	157,761.86	7680.65	4.64
Strong	152,899.32	12,543.19	7.58

cost curve can be built based on the cost-effectiveness indicators of all the possible decarbonization measures to inform policymakers and guide the prioritization of the adoption of the possible measures.

2. Improving service quality as key strategy for promoting shift from road to rail freight

Based on the result of the mode-choice estimation, it appears that improvement in service quality of rail freight, especially in safety and risk rating is a key factor that strongly drives shippers to shift towards rail transport. Improvement in time and cost of rail transports are estimated to be less effective in boosting the modal share of rail transport compared to that of risk and safety rating. This can be seen from the model coefficient for risk and safety (0.523) which indicates a relatively high positive impact on the utility of a shipper for a chosen mode. This result is consistent with the finding in earlier studies (Norojono & Young, 2003; Nugroho et al., 2015). This implies that operators of intermodal rail transport, such as PT. KAI and PT. KALOG should prioritize improvement in their service quality such that cargo damages, and loss can be minimized and streamlined, efficient services could be provided. This also underlines the importance of a certification in quality management system as an action measure that can promote the positive image of rail transport among freight forwarders and cargo owners.

3. Building sufficient capacity for rail freight transport until 2030

Currently, rail freight infrastructure on Java is underprioritized and has not been prepared to accommodate a significant increase in the demand for intermodal rail freight transport. For example, there are only a few rail terminals across the northern coastline corridors that are equipped with a proper handling equipment (such as reach stackers) to transload containers from truck to train (e.g. Sungai Lagoa (Jakarta), Klari (Karawang), Tawang (Semarang), and Kalimas (Surabaya). Furthermore, there are a number of terminals which length do not allow smooth loading and unloading operations for a train with normal number of wagons (i.e. 60 TEU). The economic benefits of transport costs reduction call for a focused investment to upgrade the capacity of rail terminals as well as rail freight network such that congestion and inefficient loading and unloading process can be avoided. This is important since rail transport will need to avoid congestion to maintain its competitive advantages (i.e. lower externalities) against road transport.

From my analysis, it can be derived that there could be an increase of 1214% (101.7 MTonne) in the volume of rail transport by 2030 under a strong intervention measure. This translates to approximately additional 7.82 MTEU (compared to 2020 level) or 23,807 TEU/day. Assuming that both Jakarta-Surabaya and Surabaya-Jakarta route would see a similar increase in volume, and that 1 locomotive could typically carry 60 TEU, this translates to a requirement of a total of 396–476 additional locomotives (taking into account the number of locomotives needed for demand fluctuations) and 5951–11,903 wagons (with single stack container freight trains). Note that these calculations assume that there will be sufficient railway capacity devoted to rail freight and could be considered as an upper limit of the additional train infrastructure needed. Nevertheless, this also means a major investment is needed to upgrade the cargo-carrying capacity of the rail freight on Java.

4. Set up a dedicated funding program for developing intermodal rail infrastructure

A dedicated funding program would help boost the development of intermodal infrastructure as part of a direct mitigation measure. One of the crucial barriers identified as a root problem for the underdeveloped intermodal infrastructure is the lack of planning perspective which is focused on intermodal freight transport. This barrier also implies that there has not been a specific focus on the development of intermodal rail infrastructure and the funding needed. Hence, a number of possible funding schemes such as public private partnership (PPP), or foreign investment and development loan can be explored. Different donor organizations with lending capacities can be approached to help establish this funding. A key prerequisite for a successful and targeted investment is a comprehensive ex-ante impact assessment aimed which includes costs benefit analysis of the relevant intermodal infrastructure developments.

5. Establish a complete, and reliable database for national freight transport flows

The availability of reliable data is indispensable to guide the development of efficient intermodal transport infrastructure and transport infrastructure in general. In the context of the intermodal transport network, a complete origin and destination data, (ideally for different modes and commodities) will be invaluable to analyze the selection of rail terminal locations and railway connections that will require a large amount of investment.

7. Contribution and future research

My contribution can be summarized as follows. Firstly, I assess the impact of the proposed modal shift measures that have been designed under consultation with the Indonesian Ministry of Transport and logistics service providers on the abatement of CO_2 emissions from freight transport across all regions on Java. This assessment has provided knowledge on the level of emissions that could be abated until 2030 using modal shift strategy with differing level of ambitions as well as its economic benefits in reducing transport costs. My results indicate that modal shift to rail freight could reduce CO_2 emissions considerably by 2030 – from 7.3% (basic ambition), 16.7% (moderate ambition) and up to 21.6% annually when the government can commit to implement strong measure. Nevertheless, a financial analysis is needed to establish the costs of implementing these measures across different ambition level and to guide the adoption of the measures based on their cost-effectiveness. Secondly, I have established a modelling framework which can be used to assess the impact of various policy intervention measures on GHG reduction from freight transport on Java. The model allows the testing of detailed abatement measures that represent "avoid", "shift", "improve approach using both a 4-step transport and emission models that are developed specifically for Java. Thirdly, I provided insights on key policy measures that could be implemented to stimulate the shift to rail freight.

There are, however, several detailed intervention measures which have not been studied which could deliver further in GHG emission savings. They are as follows:

- Other smaller corridors which may have sufficient volume to be shifted to rail have not been investigated in this study. This is due the specific scope and focus of the study, as well as limited funding available. The intervention measures are focused specifically on Northern (Jakarta- Surabaya) and the Southern (Bandung-Jombang) corridor.
- The scenarios studied do not include the potential for development of new rail terminals and connections to districts that produce high freight volumes such as in the industrial estates in Bekasi and Karawang. Currently, due to the absence of rail terminal facilities in these districts, truck mode is still used as a dominant mode of transport. Therefore, a further investigation might be needed to test the impact of building new rail terminals in these districts with adequate cargo handling equipment.

In order to better assess the benefits of the action program, several analytical and data gaps need to be addressed to obtain more robust output of the modelling. They consist of:

- Extending the SP survey to include more respondents that represent shippers, and freight forwarders from broader geographical locations such as Southern corridor of Java and commodity sectors, particularly non-containerized and bulk cargo such as palm oil, coal, fuel and chemical products.
- Experimenting with different discrete choice model types such as mixed multinomial logit model to better account for the heterogeneity in tastes and preference of shippers. Furthermore, the mode choice model can also be developed for specific commodity sectors, to better capture the unique preferences of shippers with regard to the attributes of the modes. The results of these experiment can be compared with the current model to determine model with a better fit for the shippers on Java.
- Calibrating OD data with a reliable national OD data broken down by modes. To date, a reliable national OD data and transport distances for all modes of transport are still missing. A substantial data collection, cross-validation, and correction is usually needed to generate input data with adequate quality and reliability. In my study, I noted the discrepancy between the reported OD data and my estimates, particularly on the transport from certain regions such as Jakarta, and West Java to East Java and vice versa. In the future, I aim to cross-validate my OD estimate with a reliable national OD data when it becomes available.

Funding

This research was made possible through the funding of International Climate Initiative of the Federal Ministry for the Environment, Nature Conservation and Nuclear Safety. The funding is implemented by Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) through the Indonesia component of the TRANSfer III project. The project facilitates the development of transport CO_2 mitigation actions that are ready for implementation, with a focus on freight transport in Indonesia.

Declaration of Competing Interest

The author declares no conflict of interest.

Data availability

Data will be made available on request.

Acknowledgments

Friedel Sehlleier, Lena Herliana, and Yoel Priatama (all GIZ) have provided valuable editorial assistances, comments, and feedbacks on a draft version of the report. The research was made possible through a cooperation with Ministry of Transport, particularly the AMM (Angkutan Multi Moda) unit within the DG Land Transport - a partner and beneficiary for the study in TRANSfer III programme, DG Railways, APTRINDO, and private sector stakeholders who participated in the stated preference survey. The authors are grateful for the inputs provided by La Ode Muhammad Abdul Wahid - senior researcher from BPPT (Badan Pengkajian dan Penerapan Teknologi), and the review and suggestions of Sudhir Gota in implementing the quantitative estimation framework used to estimate GHG emissions of the scenarios assessed. Furthermore, I am also appreciative of the data input provided by Zifei Yang and Zhenying Shao (all ICCT) which have been used to estimate GHG emission in this study.

Annex 1. Description of demand forecast model

The performance and the parameters of the regression model used to forecast the demand of freight transport on Java are described in the tables below. Table 1A and 1B provide the performance of the regression model. R-square value of 0.606 indicates sufficient explanatory power of the model in predicting the relationship between dependent and independent variables. The coefficient values of the model variables (i.e. GDRP, population, time) are presented in Table 1C, where all coefficients are shown to be significant. Table 1D provides coefficient values for three possible demand scenarios (i.e. low, medium, and high growth). In this study, I assumed a low growth scenario. Table 1E provides an overview of the assumed compounded annual growth rate (CAGR) for provinces on Java. This assumed CAGR forms the input for predicting transport demand volume between 2020 and 2030.

Table 1A

Regression statistics	
Multiple R	0.779
R Square	0.606
Adjusted R Square	0.606
Standard Error	0.947
Observations	13,806.000

Table 1B

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	5.000	19,059.065	3811.813	4246.430	0.000
Residual	13,800.000	12,387.585	0.898		
Total	13,805.000	31,446.651			

Table 1C

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	−17.749	0.209	−85.037	0.000
Ln GDRP (i)	−0.351	0.011	−30.743	0.000
Ln GDRP (j)	0.354	0.011	30.941	0.000
Ln Pop (i)	1.375	0.014	98.500	0.000
Ln Pop (j)	0.563	0.014	40.394	0.000
Ln Time (ij)	−0.039	0.013	−3.108	0.002

Table 1D

	Intercept	GDRP (i)	GDRP (j)	Pop (i)	Pop (j)	Time (ij)
Mid	−17.749	−0.351	0.354	1.375	0.563	−0.039
Low	−18.158	−0.374	0.331	1.347	0.536	−0.064
High	−17.340	−0.329	0.376	1.402	0.591	−0.014

Table 1E

CAGR	2015	2020	2025	2030
DKI Jakarta	5.21%	0.49%	6.93%	5.91%
Jawa Barat	5.27%	0.47%	6.61%	5.64%
Jawa Tengah	5.53%	0.54%	7.62%	6.25%
Yogyakarta	5.09%	0.46%	7.71%	6.49%
Jawa Timur	5.78%	0.50%	6.60%	5.63%
Banten	6.21%	0.54%	7.13%	6.08%

References

Akhlas, A. W. (2020). IMF projects 0.5% growth for Indonesia as global economy faces deep recession. In *The Jakarta post*. Retrieved 14 April from http://www.thejakartapost.com/news/2020/04/14/imf-projects-0-5-growth-for-indonesia-as-global-economy-faces-deep-recession.html.

Aritua, B. (2019). *The rail freight challenge for emerging economies : How to regain modal share. International development in focus*. Washington, DC: World Bank.

de Baat, M., den Hertog, V., de Jong, S., de Regt, K., & Wijgergangs, K. (2015). *Setting up a freight transportation model for Java in Indonesia*.

Ben-Akiva, M., & Bierlaire, M. (1999). Discrete choice methods and their applications to short term travel decisions. In R. Hall (Ed.), *vol. 23. Handbook of transportation science* (pp. 5–33). Springer US. https://doi.org/10.1007/978-1-4615-5203-1_2.

Bierlaire, M. (2016). *PythonBiogeme: a short introduction. Report TRANSP-OR 160706, Series on Biogeme*. Transport and Mobility Laboratory, School of Architecture, Civil and Environmental Engineering, Ecole Polytechnique Fédérale de Lausanne, Switzerland.

BPS. (2018a). *Produk Domestik Regional Bruto Provinsi-Provinsi di Indonesia Menurut Lapangan Usaha*.

BPS. (2018b). *Proyeksi Penduduk Indonesia:2015–2045 Hasil SUPAS 2015*.

CAT. (2020). *Decarbonising the Indian transport sector pathways and policies*.

Chen, S., Wu, J., & Zong, Y. (2020). The impact of the freight transport modal shift policy on China's carbon emissions reduction. *Sustainability, 12*(2), 583. https://www.mdpi.com/2071-1050/12/2/583.

Coronado, C. R., de Carvalho, J. A., & Silveira, J. L. (2009). Biodiesel CO_2 emissions: A comparison with the main fuels in the Brazilian market. *Fuel Processing Technology, 90*(2), 204–211. https://doi.org/10.1016/j.fuproc.2008.09.006

Craig, A. J., Blanco, E. E., & Sheffi, Y. (2013). Estimating the CO_2 intensity of intermodal freight transportation. *Transportation Research Part D: Transport and Environment, 22*, 49–53. https://doi.org/10.1016/j.trd.2013.02.016

DJKA. (2018). *Rencana Induk Perkeretaapian Nasional (RIPNAS) 2018*. Indonesia: Direkorat Jenderal Perkeretaapian Jakarta.

GIZ. (2021). *Technical design study of action Programme on intermodal freight transport on Java, Indonesia*.

Grütter, J. M. (2015). *MRV blueprint based on India railways NAMA. G. Consulting*.

Halim, R. A., Breemersch, T., Nahry, & Kusuma, A. (2021). *Action Programme on intermodal freight transport in Java, Indonesia (summary)*. Bonn, Germany: GIZ.

Hartono, R., Mochtar, I., & Widyastuti, H. (2017). Modelling modal shift preference of freight transport on Surabaya-Jakarta corridor : A preliminary survey. *IPTEK Journal of Proceedings Series, 3*. https://doi.org/10.12962/j23546026.y2017i6.3251

IEA. (2019). *The future of rail:Opportunities for energy and environment*.

IFEU, K., & Sgkv, K. (2009). *Comparative analysis of energy consumption and CO_2 emission of road transport and combined transport road/rail*. Geneva, Switzerland: IRU.

Indriana, I. R. (2019). *Adaptive policy decision pathways for robust shipping network investment planning in Indonesia*. Delft: Technical University Delft.

Kaack, L. H., Vaishnav, P., Morgan, M. G., Azevedo, I. L., & Rai, S. (2018). Decarbonizing intraregional freight systems with a focus on modal shift. *Environmental Research Letters, 13*(8), Article 083001. https://doi.org/10.1088/1748-9326/aad56c

Kooshian, C., Eichhorst, U., Bongardt, D., Winkelman, S., Posada, F., Yang, Z., … Novikova, V. (2017). *Compendium on greenhouse gas baselines and monitoring: Passenger and freight transport*.

Mulyono, A. T. (2019). *Perspective on future transportation research in Indonesia*.

Norojono, O., & Young, W. (2003). A stated preference freight mode choice model. *Transportation Planning and Technology, 26*(2), 1–1.

Novie, A. (2017). *Penentuan Faktor Emisi CO_2 Nasional dengan Pendekatan Analisis Bahan Bakar Minyak*. Jakarta: Temu Ilmiah XIII & Pameran Hasil Litbang ESDM.

Nugroho, M. T., Whiteing, A., & de Jong, G. (2015). *Containerised exports from Java: The impact of policies to reduce GHG emissions. The proceedings of 20th international symposium on logistics (ISL 2015) reflections on supply chain research and practice*.

Nugroho, M. T., Whiteing, A., & de Jong, G. (2016). Port and inland mode choice from the exporters' and forwarders' perspectives: Case study—Java, Indonesia. *Research in Transportation Business & Management, 19*, 73–82.

Orme, B. (1998). *Sample size issues for conjoint analysis studies*. Sawtooth Software Technical Paper,Sequim.

Ortúzar, J., & Willumsen, L. (2011). *Modelling transport* (4th ed.). John Wiley & Sons.

Pinto, J. T. D. M., Mistage, O., Bilotta, P., & Helmers, E. (2018). Road-rail intermodal freight transport as a strategy for climate change mitigation. *Environment and Development, 25*, 100–110. https://doi.org/10.1016/j.envdev.2017.07.005

PT.KAI. (2020). Spesifikasi Lokomotof CC206, Lokomotif Diesel Elektrik Terbaru PT KAI. Retrieved 2 March 2020, from https://kereta-api.info/spesifikasi-lokomotif-cc206-lokomotif-diesel-elektrik-terbaru-pt-kai-6105.htm.

Schipper, L., Marie-Lilliu, C., & Gorham, R. (1999). *Flexing the link between transport and greenhouse gas emissions-A path for the World Bank*. The World Bank.

Street, D. J., Burgess, L., & Louviere, J. J. (2005). Quick and easy choice sets: Constructing optimal and nearly optimal stated choice experiments. *International Journal of Research in Marketing, 22*(4), 459–470.

Tavasszy, L. A., & de Jong, G. (2014). *Modelling freight transport*. Elsevier.

Wanders, G. (2014). *Determining shippers' attribute preference for container transport on the Rotterdam – Venlo corridor*. Delft: Delft Univetsity of Technology.